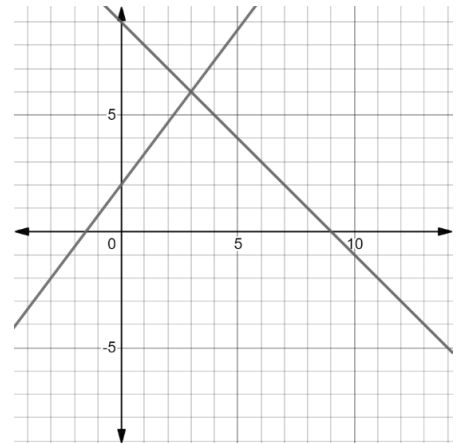


Algebra 1
Unit 5
Lesson 6 Homework

Name _____
Date _____
Period _____

The graph of the system of equations $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$ is shown.



1. To determine the solution to a system of equations, identify the

- ☐ slope.
- ☐ y-intercept.
- ☐ point of intersection.

2. Determine the solution to the system of equations from the graph.

Verify algebraically that that your solution satisfies both equations in the system.

3. $y = \frac{4}{3}x + 2$

4. $y = -x + 9$

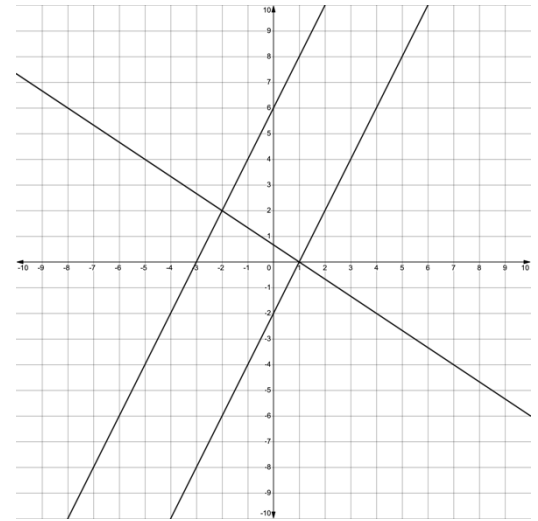
5. Choose all of the systems of equations that are equivalent to the original system of equations, $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$. Explain how you know the systems are equivalent.

$\begin{cases} -4x + 3y = 6 \\ x + y = 9 \end{cases}$	$\begin{cases} y = \frac{4}{3}x + 2 \\ 4x + 4y = 36 \end{cases}$	$\begin{cases} -\frac{4}{3}x + y = 2 \\ -x - y = -9 \end{cases}$	$\begin{cases} 8x - 6y = -2 \\ y = -x + 9 \end{cases}$
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6. Verify your choices in question 5 algebraically.

The graphs of the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ and $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ are shown.

7. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ is (____, ____). Verify the solution algebraically.



8. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ is (____, ____). Verify the solution algebraically.

9. Fill in the blanks with words from the word bank to complete the sentence.

The two systems of equations are _____ because the two systems result in _____ solution(s).	Word Bank: ☐ equivalent ☐ equal ☐ not equivalent ☐ not equal ☐ the same ☐ different
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10. Replace one equation in the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = 5 \end{cases}$ so that it is equivalent, but not equal to the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$.